

Metasploitable – NetBIOS/Samba Service Assessment

Target	Metasploitable
Scanner	NMAP
Scanner OS	Kali Linux
Scan Type	Basic Network Scan – NetBIOS/Samba
Virtualization	VMware

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- Objective:

The objective of this assessment was to identify and analyze the **NetBIOS/Samba service** running on the Metasploitable host using Nmap, including its purpose, port number, detected version, and whether the detected software appears to be current or outdated.

- Protocol Overview

NetBIOS (Network Basic Input/Output System) is a set of networking services that allow computers to communicate and share resources over a network. Also, it uses **TCP port 139**.

Samba is an open-source implementation of the **SMB** protocol, allowing non-windows OS (Unix, Linux) to seamlessly share files and printers with Windows.

- Nmap Verification

```
(kali㉿kali)-[~]
└─$ nmap -sV -p 139 192.168.31.129
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-17 12:00 EDT
Stats: 0:00:08 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 0.00% done
Nmap scan report for 192.168.31.129
Host is up (0.0014s latency).

PORT      STATE SERVICE      VERSION
139/tcp    open  netbios-ssn  Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
MAC Address: 00:0C:29:C5:A0:35 (VMware)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 14.01 seconds
```

- Nmap Results

Detected Version: Samba smbd 3.X – 4.X, Workgroup: WORKGROUP

Version Status: Outdated

Detected Version: 3.0.20-debian

Evidence: Nmap identified Samba smbd 3.X–4.X on TCP port 139, while the version installed on the Metasploitable host was verified locally as Samba 3.0.20-debian. SearchSploit identified known security issues affecting Samba 3.0.20, including a command-execution vulnerability related to the username map script.

```
(kali@kali)-[~]
$ searchsploit samba 3.0.20
```

Exploit Title	Path
Samba 3.0.10 < 3.3.5 - Format String / Security Bypass	multiple/remote/10095.txt
Samba 3.0.20 < 3.0.25rc3 - 'Username' map script' Command Execution (Metasploit)	unix/remote/16320.rb
Samba < 3.0.20 - Remote Heap Overflow	linux/remote/7701.txt
Samba < 3.6.2 (x86) - Denial of Service (PoC)	linux_x86/dos/36741.py

Shellcodes: No Results

 EXPLOIT DATABASE

Samba 3.0.20 < 3.0.25rc3 - 'Username' map script' Command Execution (Metasploit)

EDB-ID: 16320	CVE: 2007-2447	Author: METASPLOIT	Type: REMOTE	Platform: UNIX	Date: 2010-08-18
EDB Verified: ✓		Exploit: 📄 / {}		Vulnerable App:	